

```

format shortG
%clc
%clear
m = 200; % grams
W = (m/1000) * 9.81; % Newtons
%plate center
xCenter = 87.5; % mm
yCenter = 88.65; % mm
% Sensor positions (relative to center)
% These MUST be actual physical locations of the load cells
x_i = [-xCenter; xCenter; xCenter; -xCenter]; % mm
y_i = [yCenter; yCenter; -yCenter; -yCenter]; % mm
%calibration placements (relative to center)
x_k = [ 0; 50; 50; -50; -50;
       -xCenter; xCenter; xCenter; -xCenter]; % mm
y_k = [ 0; 50; -50; 50; -50;
       yCenter; yCenter; -yCenter; -yCenter]; % mm
R = [
    -23933.80,-17617.15,-19200.90,-22541.25; %(0,0)
    -22249.80,-42656.20,-16092.80,-1999.75;  %(50,50)
    -5795.30,-11834.40,-43934.10,-18822.40;  %(50,-50)
    -52045.35,-14649.90,-4935.50,-13465.55;  %(-50,50)
    -14680.15,-3754.35,-15418.60,-50675.95;  %(-50,-50)
    -69759.60,-11926.35,1640.60,-6650.70; %sensor1
    -15555.30,-66820.00,-7766.75,5753.05; %sensor2
    2614.20,-3604.45,-62894.50,-15146.15; %sensor3
    -2602.45,535.35,-10421.65,-73494.25; %sensor4
];
n=size(R,1);
B = [W*ones(n,1),W*x_k,W*y_k];
C = (R\B)';
fprintf('Condition number: %.1f\n', cond(R));
fprintf('\nfloat C[3][4] = {\n');
fprintf('    {%.6f, %.6f, %.6f, %.6f},\n', C(1,:), C(2,:), C(3,:));
fprintf('};\n');
for k = 1:n
    out = C * R(k,:)';
    W_est = out(1);
    x_est = out(2) / W_est;
    y_est = out(3) / W_est;

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```
fprintf('Point %d: W=%.3f x=%.2f (true %.2f) y=%.2f (true %.2f)\n',  
...  
      k, W_est, x_est, x_k(k), y_est, y_k(k));  
end
```